Advanced Machine Learning
Final Project

Trajectory Clustering in Vehicle Telematics Data

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Michelin R&d

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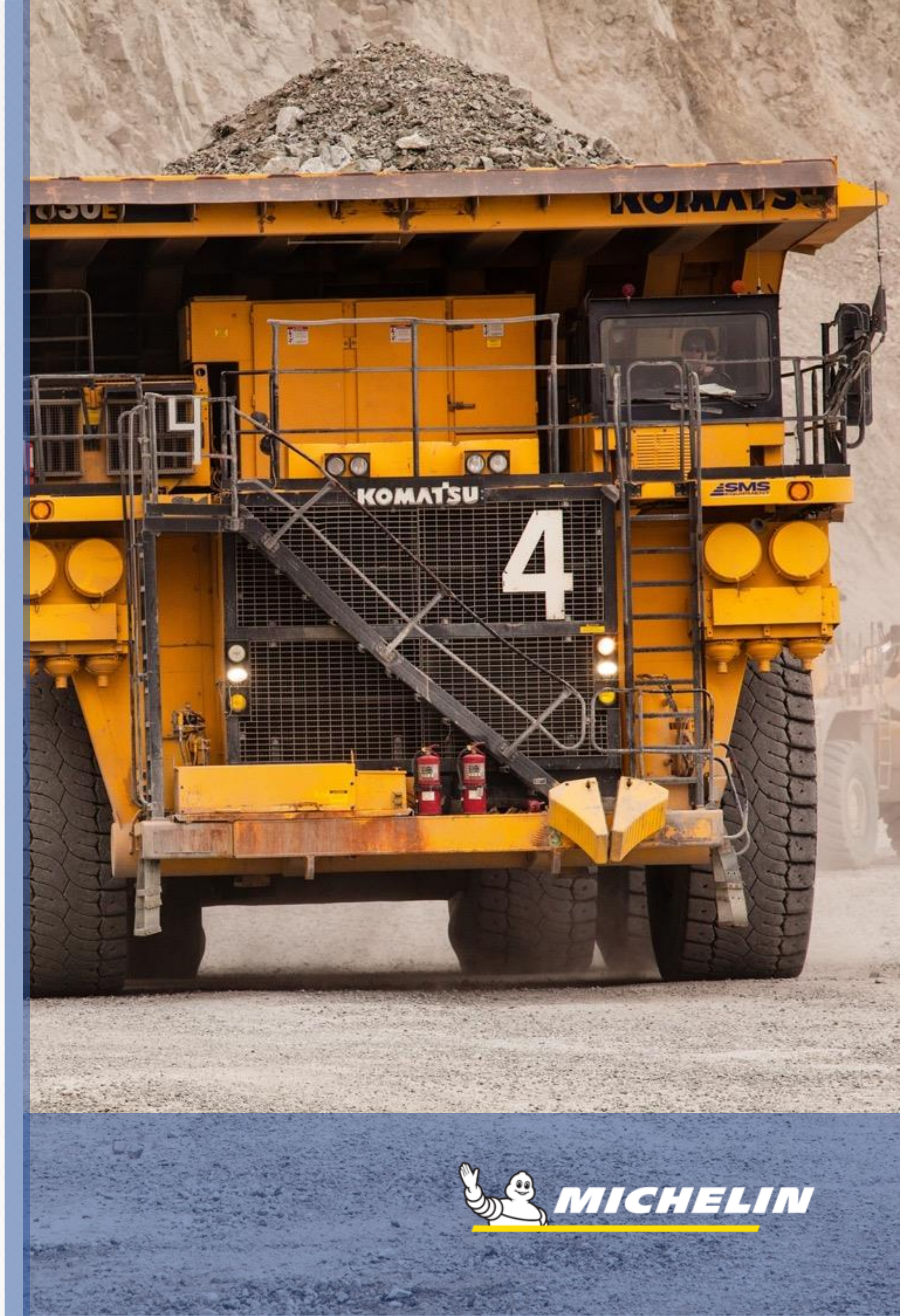
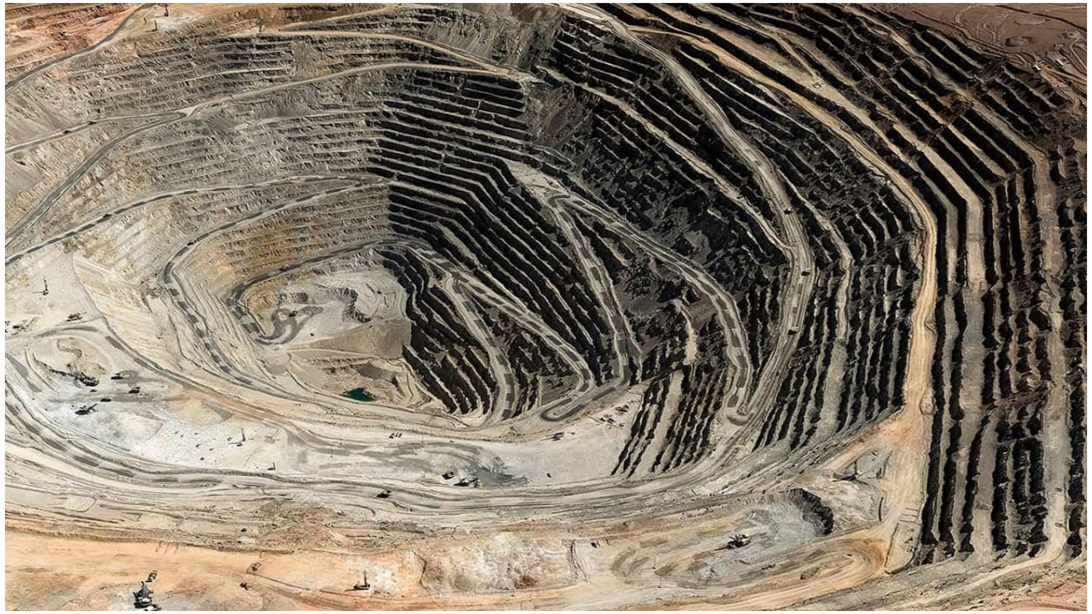


Introduction & objectives

“Our objective is to accurately and efficiently <u>cluster</u> dumpers trajectories over an extended period.”

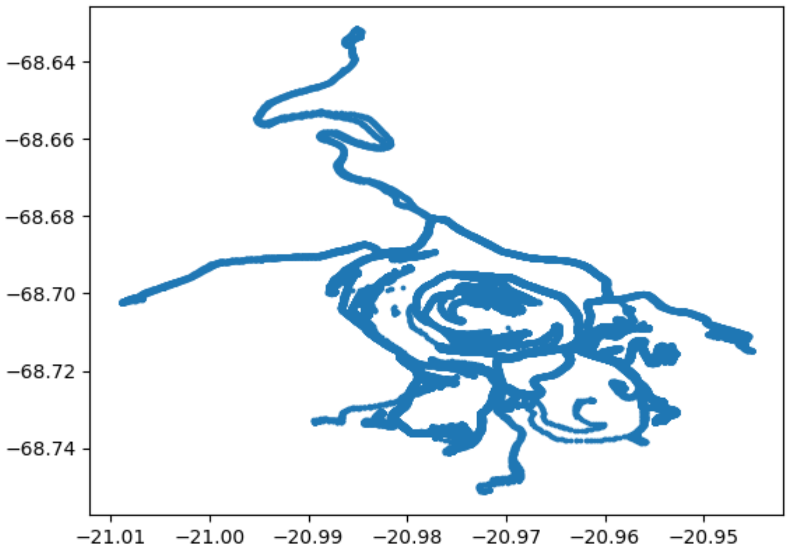


Introduction & objectives

DataSet

Mine	Collahuasi
Vehicle	CA-158
Periode	January 2023
Sensor	MEMS
Frequency	0.2 Hz

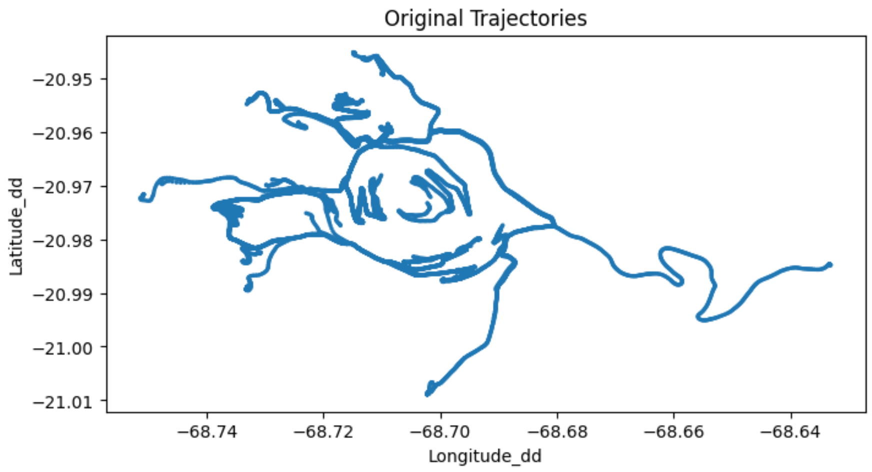
Trajectories

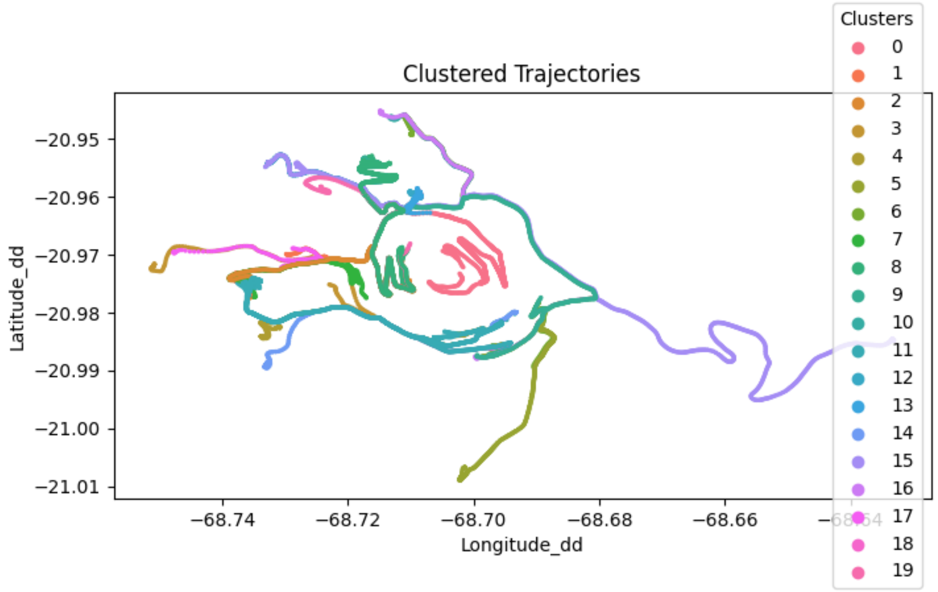


Collahuasi – Tarapacá, Chili

MICHELIN

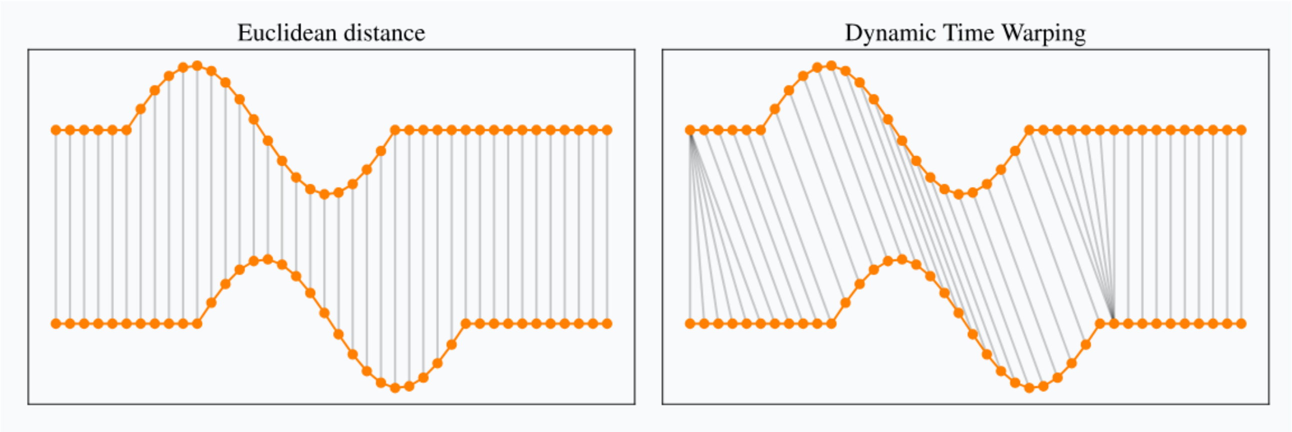
Introduction & objectives



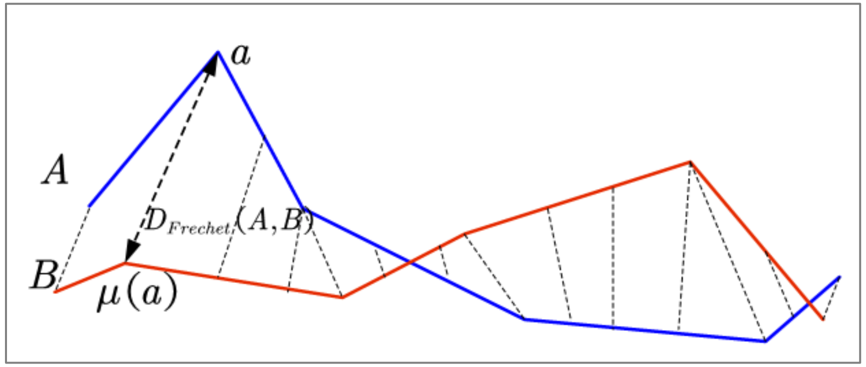


Distance measures in spatial trajectory data clustering

Euclidean Distance | Dynamic Time Warping (DTW) | Hausdorff Distance | Fréchet Distance, …

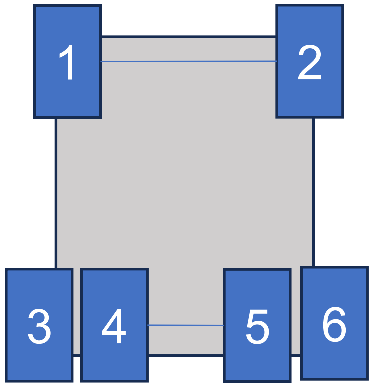


Fréchet Distance



Data Description

```
Index(['Time_utc', 'VehicleName', 'VehicleType', 'Pressure_Pa_1',
       'Pressure_Pa_2', 'Pressure_Pa_3', 'Pressure_Pa_4', 'Pressure_Pa_5',
       'Pressure_Pa_6', 'Temperature_K_1', 'Temperature_K_2',
       'Temperature_K_3', 'Temperature_K_4', 'Temperature_K_5',
       'Temperature_K_6', 'ColdPressure_Pa_1', 'ColdPressure_Pa_2',
       'ColdPressure_Pa_3', 'ColdPressure_Pa_4', 'ColdPressure_Pa_5',
       'ColdPressure_Pa_6', 'Latitude_dd', 'Longitude_dd', 'Altitude_m',
       'Speed_mps', 'epsilonX', 'epsilonY', 'epsilonZ', 'epsilonSpeed',
       'AtmosphericPressure_Pa', 'dt_s', 'dd_m', 'RatioPayload'],
      dtype='object')
```



	Time_utc	VehicleName	VehicleType	Pressure_Pa_1	Pressure_Pa_2	Pressure_Pa_3	Pressure_Pa_4	Pressure_Pa_5	Pressure_Pa_6	Temperature_K_1	...
0	2023-01-01 00:00:00.328	C-132	Dumper	993010.933333	1.003995e+06	856000.0	798005.466667	819994.533333	864994.533333	346.15	...
1	2023-01-01 00:00:05.338	C-132	Dumper	993177.933329	1.003911e+06	856000.0	798088.966665	819911.033335	864911.033335	346.15	...
2	2023-01-01 00:00:10.331	C-132	Dumper	993344.366669	1.003828e+06	856000.0	798172.183334	819827.816666	864827.816666	346.15	...
3	2023-01-01 00:00:15.325	C-132	Dumper	993510.833331	1.003745e+06	856000.0	798255.416666	819744.583334	864744.583334	346.15	...
4	2023-01-01 00:00:20.448	C-132	Dumper	993681.600000	1.003659e+06	856000.0	798340.800000	819659.200000	864659.200000	346.15	...

5 rows × 39 columns

Data Description

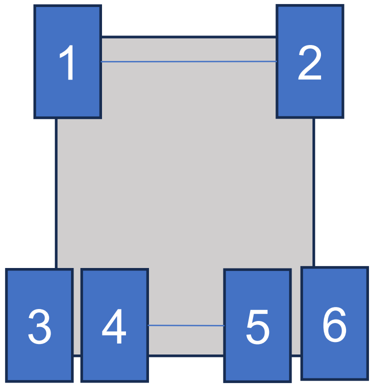
```
Index(['Time_utc', 'VehicleName', 'VehicleType', 'Pressure_Pa_1',
       'Pressure_Pa_2', 'Pressure_Pa_3', 'Pressure_Pa_4', 'Pressure_Pa_5',
       'Pressure_Pa_6', 'Temperature_K_1', 'Temperature_K_2',
       'Temperature_K_3', 'Temperature_K_4', 'Temperature_K_5',
       'Temperature_K_6', 'ColdPressure_Pa_1', 'ColdPressure_Pa_2',
       'ColdPressure_Pa_3', 'ColdPressure_Pa_4', 'ColdPressure_Pa_5',
       'ColdPressure_Pa_6', 'Latitude_dd', 'Longitude_dd', 'Altitude_m',
       'Speed_mps', 'epsilonX', 'epsilonY', 'epsilonZ', 'epsilonSpeed',
       'AtmosphericPressure_Pa', 'dt_s', 'dd_m', 'RatioPayload'],
      dtype='object')
```

```
df['RatioPayload']

0         0.0
1         0.0
2         0.0
3         0.0
4         0.0
          ...
201307    1.0
201308    1.0
201309    1.0
201310    1.0
201311    1.0
```

Empty

Loaded



	Time_utc	VehicleName	VehicleType	Pressure_Pa_1	Pressure_Pa_2	Pressure_Pa_3	Pressure_Pa_4	Pressure_Pa_5	Pressure_Pa_6	Temperature_K_1	...
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4	2023-01-01 00:00:20.448	C-132	Dumper	993681.600000	1.003659e+06	856000.0	798340.800000	819659.200000	864659.200000	346.15	...

5 rows × 39 columns

RESOURCES

https://github.com/mohamedELBAHA/5AIMDS